COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07-09-26	**DATE SUBMITTED:**	07-09-26
NAME:	Masongsong, John Marvin B.	**INSTRUCTOR:**	Engr. Jimlord Quejado

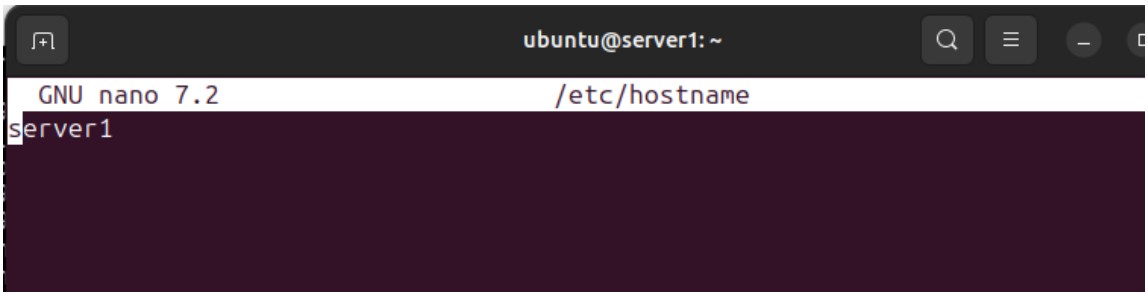
ACTIVITY NO. 1.1 **Creating Virtual Machines**

1. PROCEDURE OUTPUT

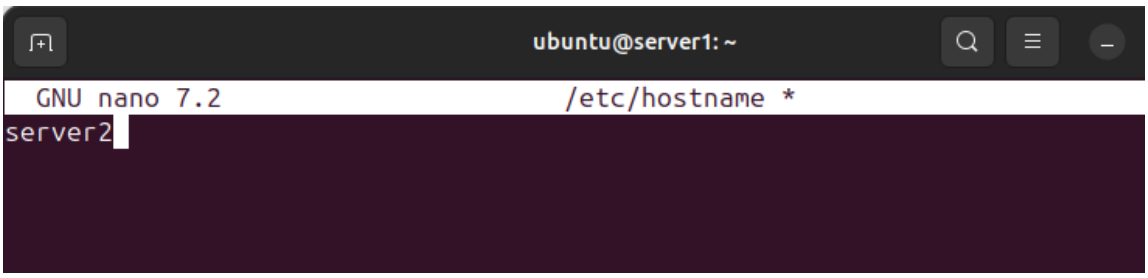
Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command sudo nano /etc/hostname

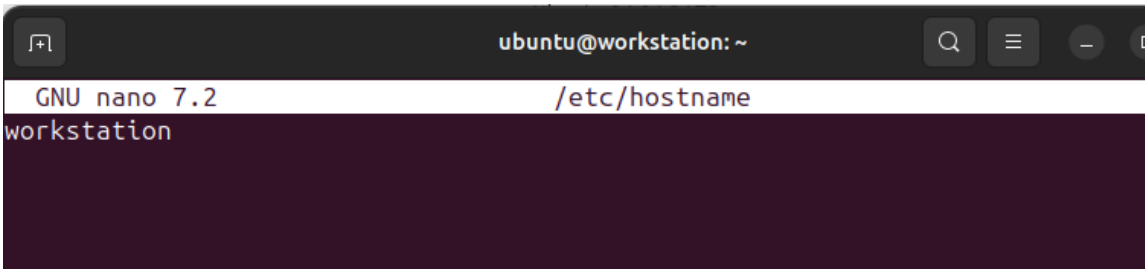
1.1 Use server1 for Server 1



1.2 Use server2 for Server 2

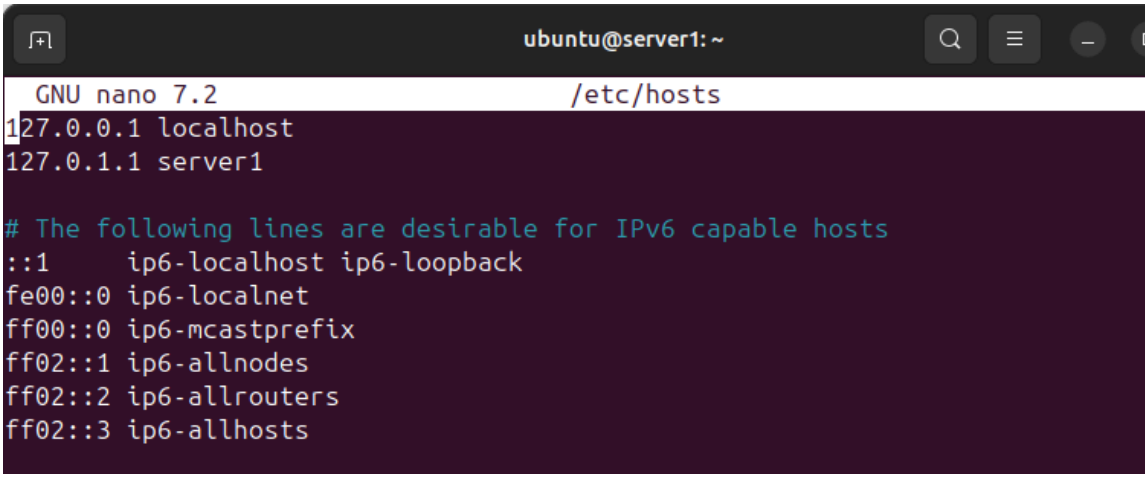


1.3 Use workstation for the Local Machine

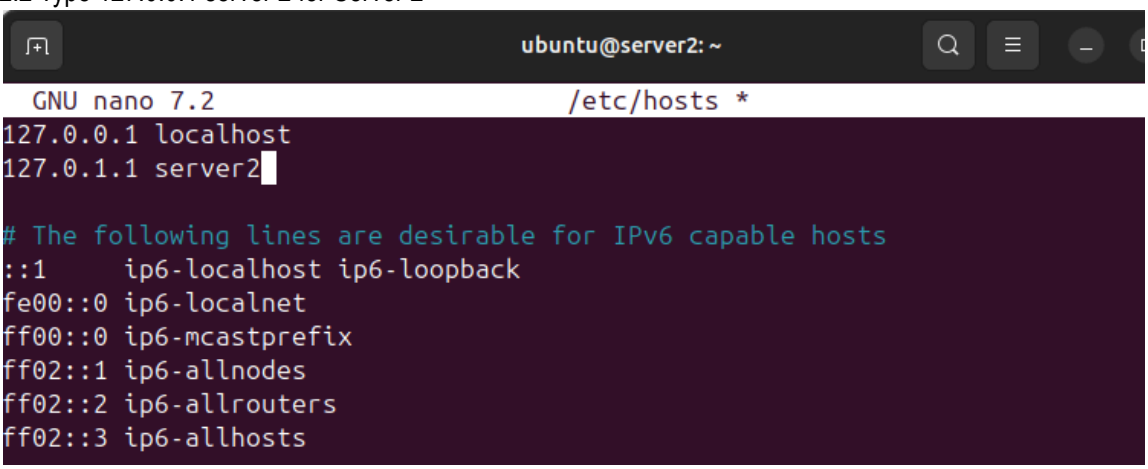


2. Edit the hosts using the command sudo nano /etc/hosts. Edit the second line.

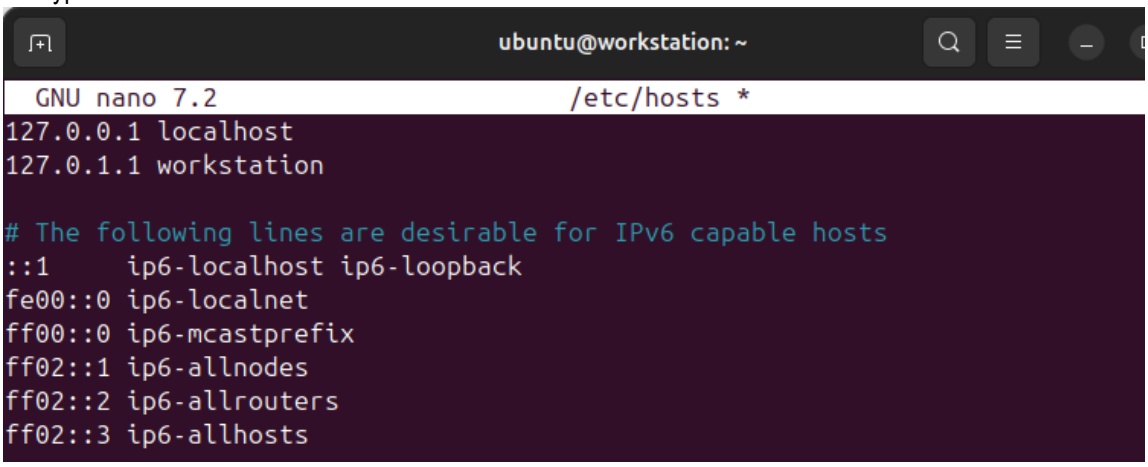
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2

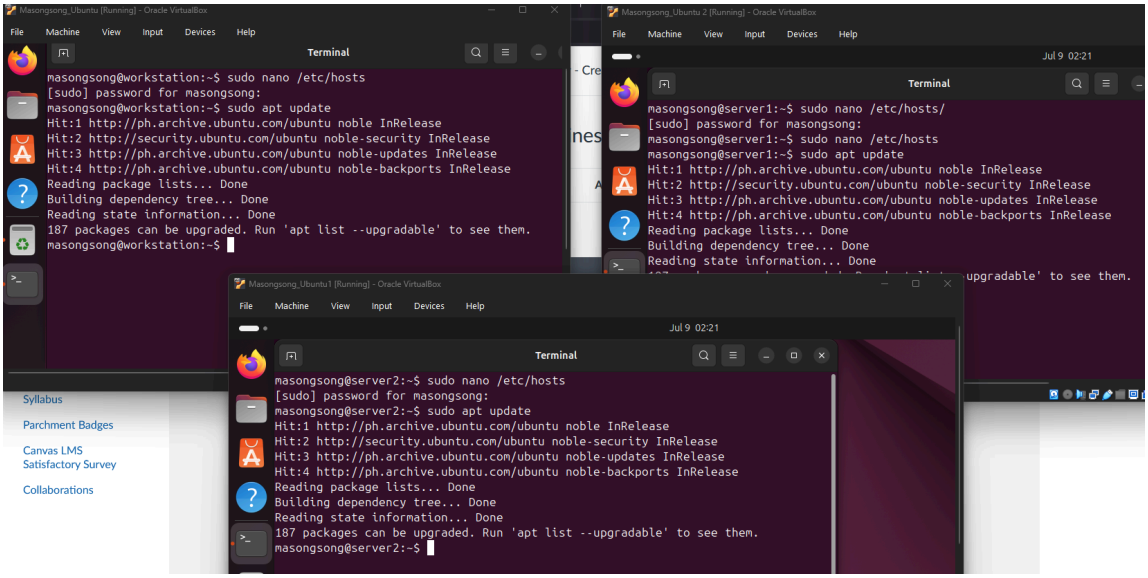


2.3 Type 127.0.0.1 workstation for the Local Machine

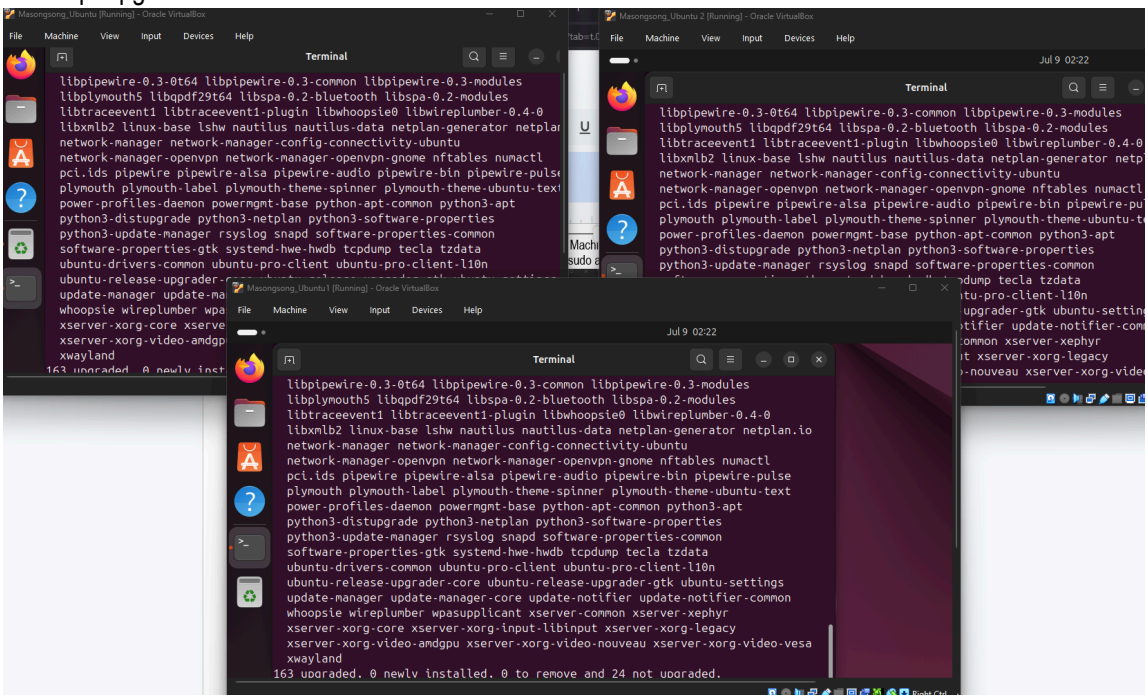


Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

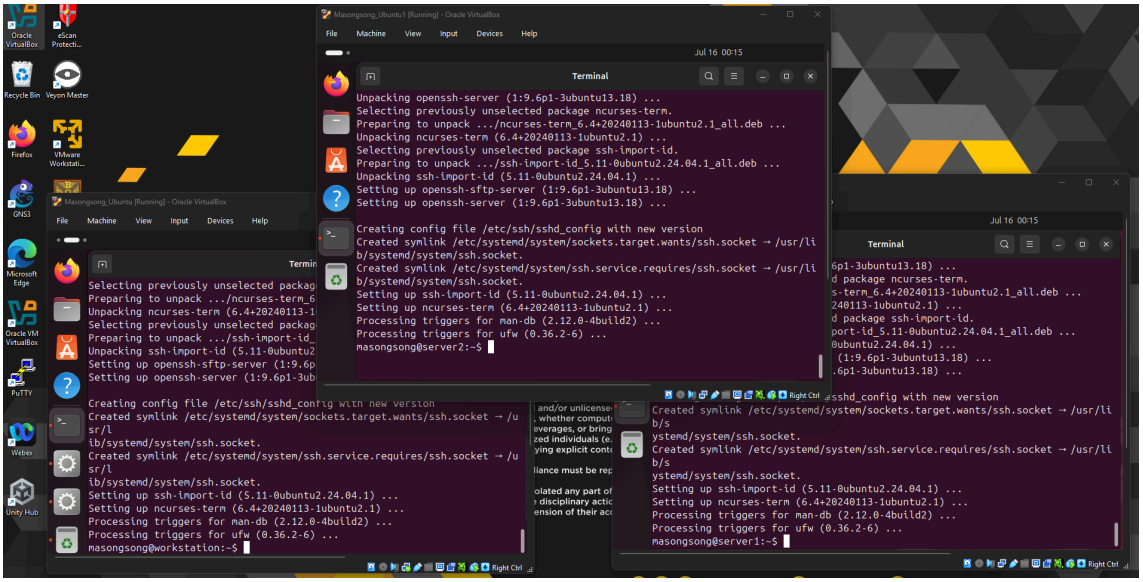
1. Upgrade the packages by issuing the command sudo apt update and sudo apt upgrade respectively.
-sudo apt update



-sudo apt upgrade



2. Install the SSH server using the command sudo apt install openssh-server.



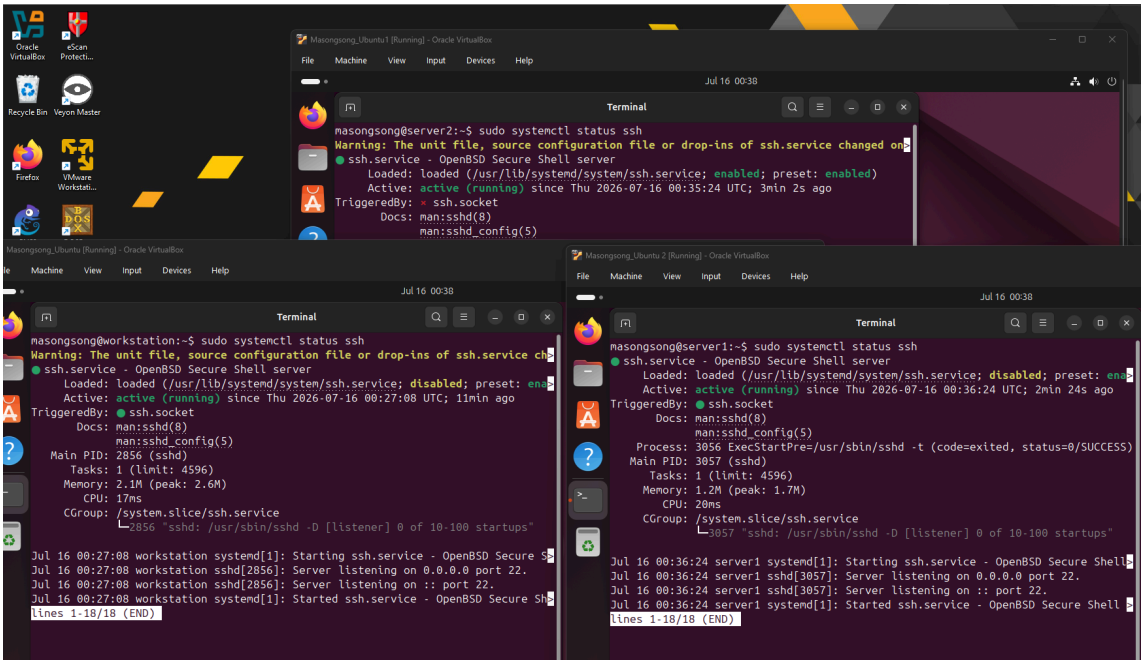
3. Verify if the SSH service has started by issuing the following commands:
   3.1 sudo service ssh start

```
masongsong@workstation:~$ sudo service ssh start
[sudo] password for masongsong:
```

```
masongsong@server1:~$ sudo service ssh start
masongsong@server1:~$
```

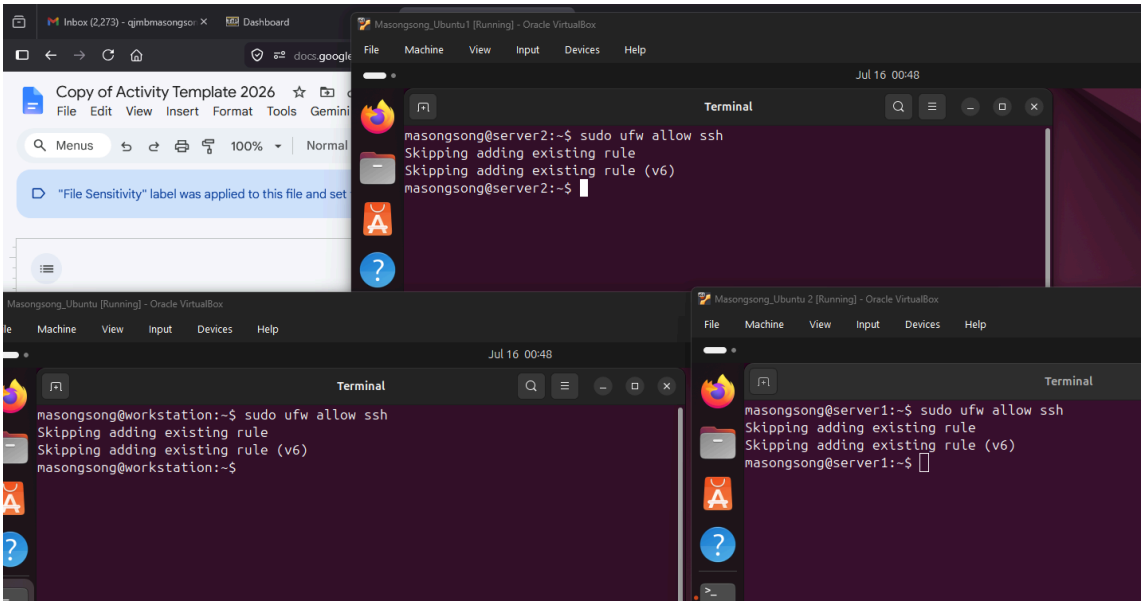
```
masongsong@server2:~$ sudo service ssh start
masongsong@server2:~$
```

   3.2 sudo systemctl status ssh

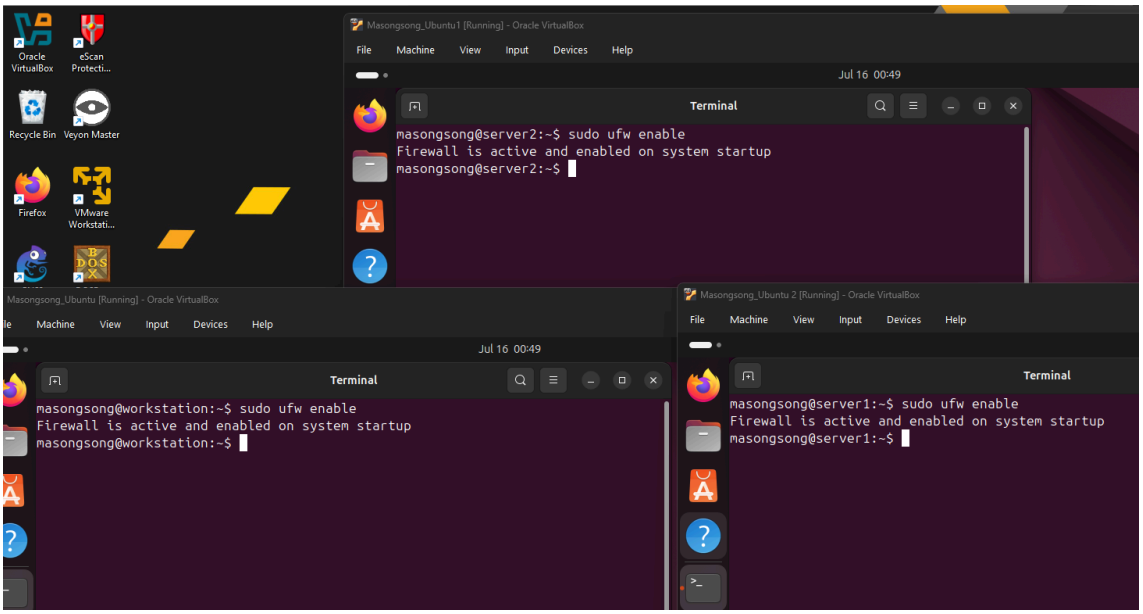


4. Configure the firewall to all port 22 by issuing the following commands:

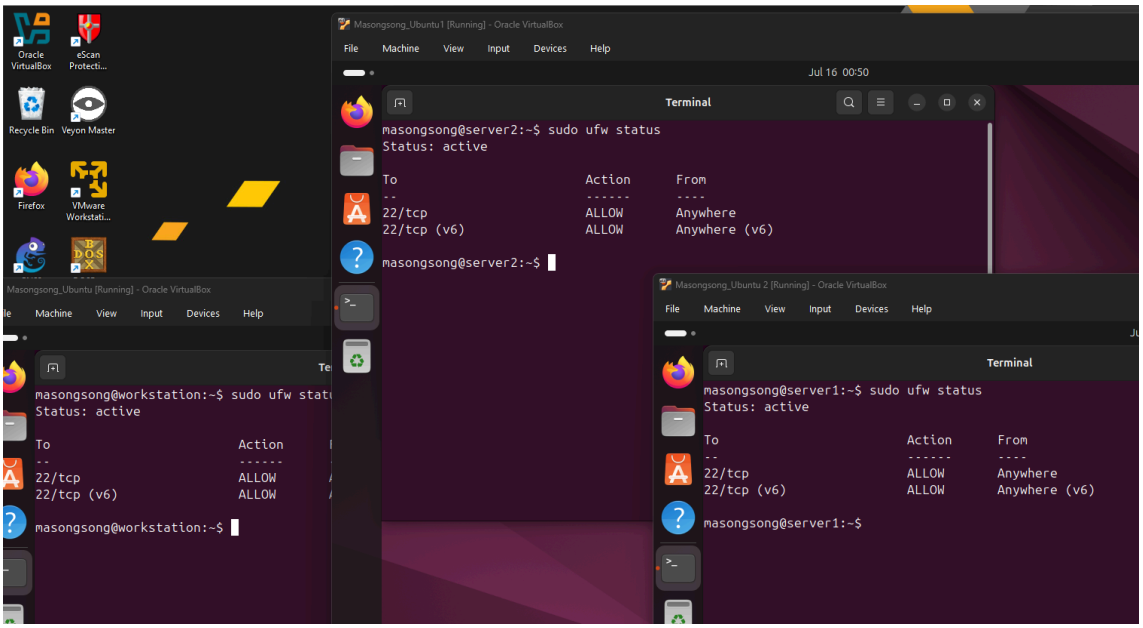
4.1 sudo ufw allow ssh



4.2 sudo ufw enable



4.3 sudo ufw status



Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command ifconfig and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
   1.1 Server 1 IP address: **192.168.56.127**
   1.2 Server 2 IP address: **192.168.56.128**
   1.3 Server 3 IP address: **192.168.56.129**

2. Make sure they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: Successful

```
masongsong@workstation:~$ ping 192.168.56.128
PING 192.168.56.128 (192.168.56.128) 56(84) bytes of data.
64 bytes from 192.168.56.128: icmp_seq=1 ttl=64 time=2.25 ms
64 bytes from 192.168.56.128: icmp_seq=2 ttl=64 time=0.380 ms
64 bytes from 192.168.56.128: icmp_seq=3 ttl=64 time=1.16 ms
^C
--- 192.168.56.128 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2396ms
rtt min/avg/max/mdev = 0.380/1.262/2.251/0.767 ms
masongsong@workstation:~$
```

2.2 Connectivity test for Local Machine 1 to Server 2: Successful

```
masongsong@workstation:~$ ping 192.168.56.129
PING 192.168.56.129 (192.168.56.129) 56(84) bytes of data.
64 bytes from 192.168.56.129: icmp_seq=1 ttl=64 time=1.81 ms
64 bytes from 192.168.56.129: icmp_seq=2 ttl=64 time=1.40 ms
64 bytes from 192.168.56.129: icmp_seq=3 ttl=64 time=0.795 ms
^C
--- 192.168.56.129 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2009ms
rtt min/avg/max/mdev = 0.795/1.334/1.813/0.417 ms
masongsong@workstation:~$
```

2.3 Connectivity test for Server 1 to Server 2: ☐ Successful ☐ Not Successful

```
masongsong@server1:~$ ping 192.168.56.129
PING 192.168.56.129 (192.168.56.129) 56(84) bytes of data.
64 bytes from 192.168.56.129: icmp_seq=1 ttl=64 time=0.054 ms
64 bytes from 192.168.56.129: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 192.168.56.129: icmp_seq=3 ttl=64 time=0.043 ms
64 bytes from 192.168.56.129: icmp_seq=4 ttl=64 time=0.062 ms
64 bytes from 192.168.56.129: icmp_seq=5 ttl=64 time=0.031 ms
^C
--- 192.168.56.129 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4894ms
rtt min/avg/max/mdev = 0.031/0.048/0.062/0.010 ms
masongsong@server1:~$
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the local machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, ssh jvtaylar@192.168.56.120

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, jvtaylar@server1

```
masongsong@workstation:~$ ssh masongsong@192.168.56.128
The authenticity of host '192.168.56.128 (192.168.56.128)' can't be established.
ED25519 key fingerprint is SHA256:A9gs3TwX9Ghisrl0PsdpH72nheeWSbUsjYmuyGA85zE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.56.128' (ED25519) to the list of known hosts
.
masongsong@192.168.56.128's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted
applicable law.

masongsong@server2:~$
```

2. Logout of Server 1 by issuing the command control + D.

```
masongsong@server2:~$
logout
Connection to 192.168.56.128 closed.
masongsong@workstation:~$
```

3. Do the same for server 2

```
Terminal
masongsong@workstation:~$ ssh masongsong@192.168.56.128
masongsong@192.168.56.128's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Thu Jul 16 01:01:12 2026 from 192.168.56.127
masongsong@server2:~$
logout
Connection to 192.168.56.128 closed.
masongsong@workstation:~$
```

4. Edit the hosts of the Local Machine by issuing the command sudo nano /etc/hosts. Below all texts type the following:
   4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)
   4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)
   4.3 Save the file and exit.

```
  GNU nano 7.2                          /etc/hosts *
127.0.0.1 localhost
127.0.1.1 workstation
192.168.56.128 server1
192.168.56.129 server2

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

Save modified buffer?
 Y Yes
 N No           ^C Cancel
```

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do ssh jvtaylar@server1. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

```
masongsong@workstation:~$ ssh masongsong@server1
The authenticity of host 'server1 (192.168.56.128)' can't be established.
ED25519 key fingerprint is SHA256:A9gs3TwX9Ghisrl0PsdpH72nheeWSbUsjYmuyGA8
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
1 updates could not be installed automatically. For more deta
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Thu Jul 16 01:07:57 2026 from 192.168.56.127
masongsong@server1:~$
```

```
masongsong@workstation:~$ ssh masongsong@server2
The authenticity of host 'server2 (192.168.56.129)' can't be established.
ED25519 key fingerprint is SHA256:5UiLCTBmB2R+AHnXSY0UJhQK59Dxr6mM1y9F8hC
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
1 updates could not be installed automatically. For more detai
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Thu Jul 16 01:04:50 2026 from 192.168.56.127
masongsong@server2:~$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. How are we able to use the hostname instead of IP address in SSH commands?
We are able to use the hostname by editing /etc/hosts and inputting the ip addresses of server 1 and server 2 plus their hostname right below the workshop's hostname. After that we are now able to just input the hostname instead of the ip address which can take a long time.

2. How secured is SSH?
SSH is pretty secure if you are going to use it for connections and networking, since it uses public and private keys, you are able to remotely connect to certain ip addresses and configure it from where you are. But, there are some downsides like SSH being able to be opened through brute forcing.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar corrections and explaining of errors**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used AI to explain errors on why it does not work as it was intended, after I asked to help out on my grammar for analysis and conclusions.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.